

Lab 4 — Decision Tree using scikit-learn with Visualisation

Aim

To build and visualise a Decision Tree Classifier using scikit-learn on the Iris dataset.

Algorithm / Procedure

1. Load the Iris dataset using `load_iris()`.
2. Separate the dataset into features and target classes.
3. Split the data into training and testing sets.
4. Create a `DecisionTreeClassifier` using the Entropy criterion.
5. Train the classifier using the training data.
6. Predict the classes of the test data.
7. Calculate the classification accuracy.
8. Visualise the trained decision tree using `plot_tree()`.
9. Save and display the decision tree image.

Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)
```

```
# Create and train Decision Tree
clf = DecisionTreeClassifier(
    criterion='entropy',
    random_state=1
)

clf.fit(X_train, y_train)

# Prediction
y_pred = clf.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

# Visualise Decision Tree
plt.figure(figsize=(12, 8))

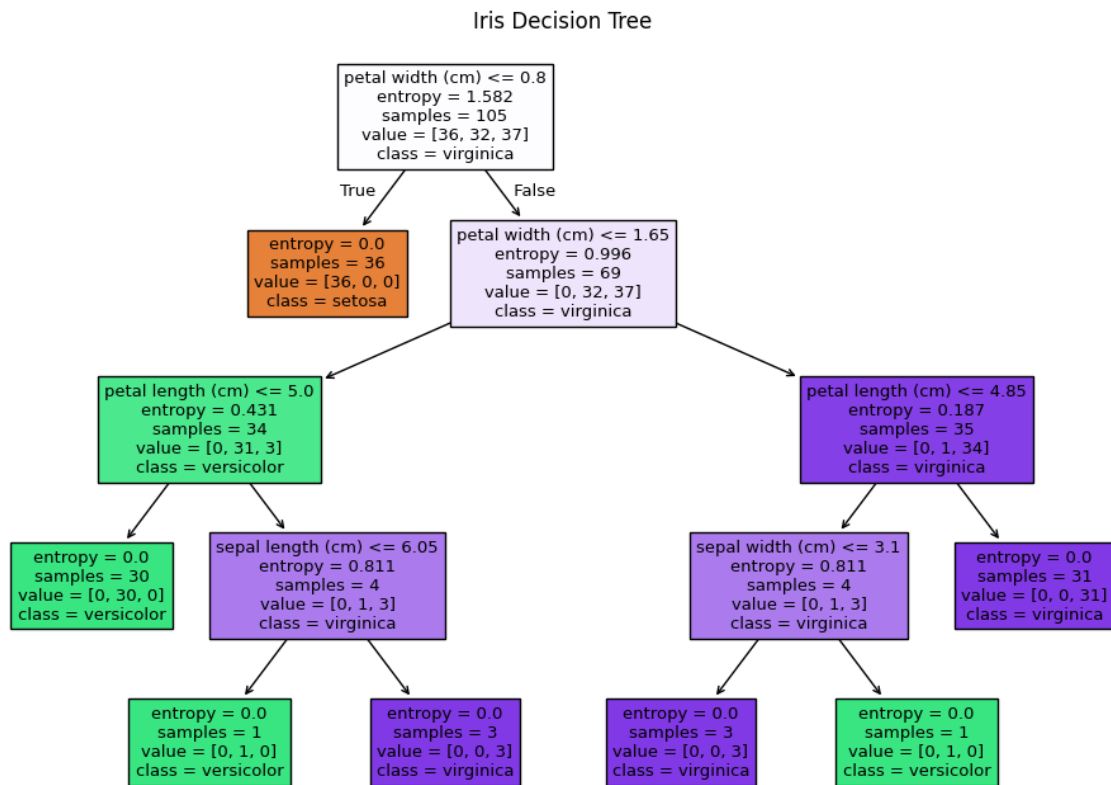
plot_tree(
    clf,
    feature_names=iris.feature_names,
    class_names=iris.target_names,
    filled=True
)

plt.title("Iris Decision Tree")
plt.savefig("iris_tree.png")
plt.show()

print("Decision tree saved as iris_tree.png")
```

OUTPUT

Accuracy: 0.9555555555555556



Result

The Decision Tree Classifier was successfully implemented using scikit-learn and visualised using Matplotlib. The model achieved approximately 95.5% accuracy on the Iris test dataset.